

- By Theorem 3, the matrix equation $Ax=0$ has the same solution as the system of equations corresponding to the augmented matrix $[A \ 0]$.
 - By row reduction algorithm and Theorem 1, the augmented matrix can be converted to a unique rref.
 - By Theorem 2, the matrix A has no free variables. $\therefore$ By defn, both a_1 and a_2 are pivot columns.
 - Suppose, Row 1 has the pivot in Column 1. $\therefore$ Row 1 Column 1 = $\square$
 - From Cond'n 2 of the matrix to be in echelon form, Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
 - Suppose, Row 2 has no pivots. $\therefore$ Row 2 Column 2 = 0
 - $\therefore$ Col 2 is a pivot column, either it will be present in Row 3 or Row 4.
 - This will violate Cond'n 1. of the matrix to be in echelon form.
 - $\therefore$ Row 2 Column 2 = $\square \rightarrow$ pivot
 - By Cond'n 3 of the matrix to be in echelon form, Row 3 Column 2 = Row 4 Column 2 = 0.
 - $\therefore$ There is no constraint, Row 1 Column 2 = *
- Final echelon form of $A = \begin{bmatrix} \square & * \\ 0 & \square \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

- Suppose, Row 1 has the pivot in Column 2. Since Column 1 is a pivot column, there is a pivot in atleast one of Row 2 to 4. This is a contradiction to Cond'n. 2 of the matrix to be in echelon form.
 - $\therefore$ ~~Row 1~~ Row 1 can't have pivot in Column 2. $\square$
- 26) A is a 4×3 matrix, $A = [a_1 \ a_2 \ a_3]$, such that $\{a_1, a_2\}$ is linearly independent and a_3 is not in $\text{Span}\{a_1, a_2\}$.
- Soln: $\therefore \{a_1, a_2\}$ are linearly independent, by defn, $c_1 a_1 + c_2 a_2 = 0$ if and only if $c_1 = c_2 = 0$.
- $\therefore a_3$ is not in $\text{Span}\{a_1, a_2\} \therefore$ By defn, a_3 is not a linear combination of a_1 and a_2 . $\therefore$ For all c_1 and $c_2 \in \mathbb{R}$, $c_1 a_1 + c_2 a_2 \neq a_3$.
- Suppose, $\exists x_1, x_2, x_3 \in \mathbb{R}$ s.t. $x_1 a_1 + x_2 a_2 + x_3 a_3 = 0$.